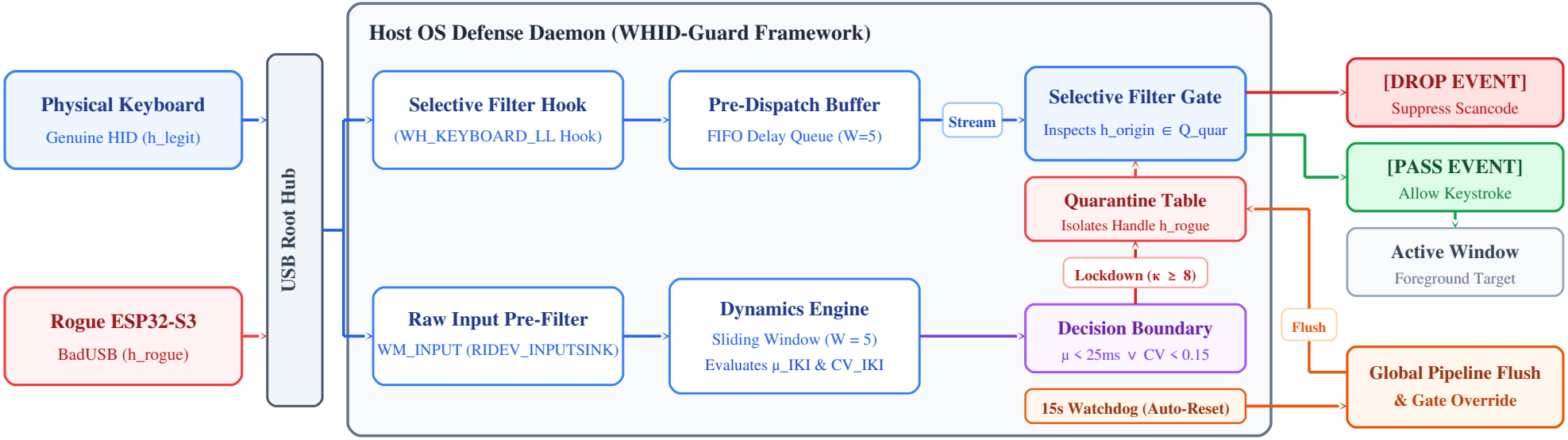

Host OS Defense Daemon (WHID-Guard Framework)
Physical Keyboard
Genuine HID (h_legit)
Rogue ESP32-S3
BadUSB (h_rogue)
USB Root Hub
Selective Filter Hook
(WH_KEYBOARD_LL Hook)
Raw Input Pre-Filter
WM_INPUT (RIDEV_INPUTSINK)
Pre-Dispatch Buffer
FIFO Delay Queue (W=5)
Dynamics Engine
Sliding Window (W = 5)
Evaluates μ_IKI & CV_IKI
Stream
Selective Filter Gate
Inspects h_origin ∈ Q_quar
Quarantine Table
Isolates Handle h_rogue
Lockdown (κ ≥ 8)
Decision Boundary
μ < 25ms ∨ CV < 0.15
15s Watchdog (Auto-Reset)
[DROP EVENT]
Suppress Scancode
[PASS EVENT]
Allow Keystroke
Active Window
Foreground Target
Flush
Global Pipeline Flush
& Gate Override